

BDPH FINAL MASTER ENTERPRISE SCOPE OF WORK

Transport ERP + Intelligent GPS Tracking + Enterprise Operations Ecosystem

This document defines the complete enterprise-level development scope for the BDPH Enterprise Platform. The objective is to create a scalable, modular, cloud-native, future-ready operational ecosystem integrating: • GPS Tracking • Transport ERP • Fleet Operations • Inventory Management • Document Management • Maintenance Systems • Billing & Accounts • Analytics & AI • IoT Ecosystem • Enterprise Reporting • Mobile Applications • Future Smart Enterprise Expansion This document is designed to avoid confusion between BDPH and the software development company by clearly defining: • Modules • Features • Development phases • Future scalability • Deliverables • Technical expectations • Expansion readiness

1. Platform Architecture Philosophy

The platform must be developed using modular enterprise architecture. Every module should work independently while remaining fully integrated with the central ecosystem. The architecture must support: • API-driven integrations • Future ERP expansion • Cloud-native deployment • Mobile applications • AI integrations • Industrial IoT integrations • Multi-branch operations • Enterprise scalability • Third-party integrations • Future smart automation systems The development company must ensure future modules can be added without rebuilding the core system.

2. Project Development Roadmap

Phase	Scope	Expected Outcome
Phase 1	Core Platform + Login + Dashboard + User Management	Enterprise foundation ready
Phase 2	GPS + Vehicles + Drivers + Maps	Tracking ecosystem operational
Phase 3	Trips + Fuel + Tyres +Battery+ Maintenance	Transport operations ecosystem ready
Phase 4	Billing + Accounts + Inventory + Documents	ERP operational layer ready
Phase 5	Reports + Analytics + AI + IoT + Alerts	Enterprise intelligence ecosystem ready
Phase 6	Mobile Apps + ERP Integrations + Future Expansion	Scalable future-ready ecosystem

3. Core Enterprise Platform

Purpose: Central operational control layer for all enterprise activities. Features: • Secure login system • Role-based access control • Multi-user management • Multi-branch support • Admin dashboard • Real-time operational dashboard • Dark/light UI themes • User activity logs • Notification center • Cloud deployment readiness • Mobile responsive interface

4. GPS Tracking & Intelligent Telematics

Purpose: Real-time fleet monitoring and live operational visibility. Features: • Live GPS tracking • Live map monitoring • Vehicle ignition tracking • Speed monitoring • Idle monitoring • Offline detection • Overspeed alerts • Geo-fencing • Trip

• playback • Route replay • Route optimization • Fleet heatmaps • Live operational dashboard • Real-time vehicle health monitoring

5. Vehicle & Asset Management

Purpose: Centralized management of vehicles, compliance, and operational assets. Features: • Vehicle master database • RC management • Insurance tracking • Permit management • Fitness tracking • Document expiry alerts • Vehicle capacity tracking • Service schedules • Asset tagging • Asset lifecycle management

6. Driver Management

Purpose: Complete operational management of drivers. Features: • Driver profiles • License tracking • Driver assignment • Driver attendance • Shift management • Driver behavior analytics • Performance scoring • Violation tracking • Driver mobile readiness

7. Trips & Operations Management

Purpose: Transport operations planning and execution ecosystem. Features: • Trip creation • Dispatch management • Route assignment • Loading/unloading points • Trip status tracking • Trip profitability analysis • Fleet scheduling • Workflow automation • Operational tracking dashboards

8. Fuel & AdBlue Management

Purpose: Fuel monitoring and operational efficiency management. Features: • Manual fuel entries • Fuel mileage calculations • Fuel theft detection logic • AdBlue tracking • Refill management • Fuel analytics • Consumption trend analysis • Future sensor integration support

9. Tyre Management

Purpose: Complete tyre lifecycle and operational management. Features: • Tyre allocation • Tyre position tracking • Tyre lifecycle management • Retreading tracking • Tyre replacement history • Tyre performance analytics • Tyre inventory integration

10. Breakdown & Maintenance Management

Purpose: Preventive and corrective maintenance ecosystem. Features: • Breakdown entries • Repair history • Workshop management • Preventive maintenance schedules • Maintenance alerts • Downtime analysis • Spare parts tracking • Predictive maintenance AI readiness

11. Billing & Accounts

Purpose: Financial and operational accounting ecosystem. Features: • Trip-based billing • Customer ledger • Expense management • Fuel expense tracking • Driver advance tracking • Toll management • Repair expenses • Invoice generation • GST-ready structure • Vehicle profitability analytics • Trip profitability analytics

12. Inventory Management

Purpose: Warehouse and stock management ecosystem. Features: • Spare parts inventory • Tyre inventory • Lubricants inventory • Warehouse management • Stock monitoring • Low stock alerts • Barcode & QR integration • Vendor management • Purchase workflows • Inventory audit logs

13. Enterprise Document Management

Purpose: Centralized enterprise document ecosystem. Features: • Digital document repository • Vehicle document storage • Invoice & contract storage • Role-based document access • Version control • Approval workflows • Cloud synchronization • Compliance document tracking • Document expiry reminders

14. Reports & Analytics

Purpose: Executive and operational intelligence dashboards. Features: • Daily vehicle reports • Fuel reports • Trip reports • Idle reports • Profitability reports • Executive dashboards • Interactive charts • KPI analytics • PDF/Excel export • Business intelligence compatibility

15. AI & Smart Intelligence

Purpose: Future-ready AI operational intelligence ecosystem. Features: • Predictive maintenance AI • Fuel optimization AI • Driver risk scoring • AI anomaly detection • Machine learning analytics • Smart operational recommendations • Natural language assistant

16. IoT & Sensor Ecosystem

Purpose: Industrial smart device and sensor integration layer. Features: • Fuel sensors • Temperature and Pressure sensors • CAN bus integration • RFID integration • Dashcam support • Industrial IoT compatibility • Edge computing readiness

17. Mobile Applications & Future Expansion

Purpose: Future enterprise expansion ecosystem. Features: • Android application readiness • iOS application readiness • Progressive Web App • Offline synchronization • Push notifications • ERP integration readiness • Blockchain readiness • Digital twin readiness • Smart city ecosystem compatibility

18. Enterprise Security, Cloud & DevOps

Purpose: Enterprise-grade operational reliability and scalability. Features: • Cloud-native deployment • Docker/Kubernetes readiness • Auto-scaling architecture • CI/CD pipelines • Multi-factor authentication • API security • Backup & disaster recovery • Monitoring & observability • Audit logs

19. Battery Management System

Purpose: Battery monitoring and Electrical health management ecosystem

Features: Battery voltage monitoring, Battery Health Tracking, Charging/Discharging monitoring, Battery replacement history, Low battery alerts, Battery Analytics dashboard, vehicle wise battery monitoring, Battery life cycle analytics, critical battery notifications, Future sensor integration support.

20. Developer Tracking Matrix

Module	Frontend	Backend	API	Testing	Deployment
GPS Tracking					
Vehicles & Drivers					
Trips & Operations					
Fuel & AdBlue					
Tyres					
Battery					
Maintenance					
Billing & Accounts					
Inventory					
Document Management					
Reports & Analytics					
AI & IoT					

21. Future Scalability & Expansion Requirements

The architecture must remain modular and API-driven so future enterprise modules can be added without changing the core platform. Future implementation possibilities: • HR & Payroll systems • Smart warehouse systems • AI operational automation • EV fleet monitoring • Drone operations monitoring • Blockchain audit systems • Autonomous fleet systems • Industrial IoT ecosystems • Smart city integrations • Advanced BI dashboards • Vendor portals • Customer portals • Technician mobile apps • Predictive analytics engines • Advanced AI copilots

22. Final Deliverables

Deliverable	Included
Frontend Enterprise Platform	Yes
Backend APIs	Yes
Database Architecture	Yes
GPS Tracking Engine	Yes
Transport ERP Modules	Yes
Inventory Management	Yes
Document Management	Yes
Billing & Accounts	Yes
Analytics Dashboards	Yes
Cloud Deployment	Yes
Testing & QA	Yes
Technical Documentation	Yes
Admin Dashboard	Yes

Final Enterprise Vision:
The BDPH platform shall evolve into a unified enterprise operational ecosystem integrating transport ERP, intelligent

telematics, AI analytics, maintenance systems, inventory management, billing, document management, industrial IoT systems, and future enterprise technologies within a scalable cloud-native architecture. The development company must ensure modular implementation standards so future enterprise modules can be added easily without rebuilding the platform.